

Supplementary Material

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Abstract

In the supplementary material, we include background information on the data set, method details of the reference-panel construction and the LASSO model, data-filtering strategies, and the rationale behind the choice of parameters.

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S1 Data and reference panel

S1.1 Pf3k data set

The Pf3k (Pf3k Consortium, 2016) data set includes 2,640 *Plasmodium falciparum* samples.

These can be divided into field and lab samples based on their sources, and into clonal and mixed samples based on the mixtures identified by our work (Zhu et al., 2018, 2019); see Table S1.1 for details. Note that we drop the duplicated samples and use only the preferred ones.

We also exclude samples where the sequencing depth is below 30 and where less than 30% of sites are callable.

	Clonal samples	Mixed samples	Total
Field samples	1365	979	2344
Lab samples	86	27	113
Total	1451	1006	2457

Table S1.1: Field samples are collected from multiple sampling sites in Africa and Asia (Duffy et al., 2015; Kamau et al., 2015; MalariaGEN *Plasmodium falciparum* Community Project, 2016; Mobegi et al., 2014; Amato et al., 2017; Miotto et al., 2013; Ocholla et al., 2014; Amambua-Ngwa et al., 2012; Wong et al., 2017). Clonal and mixed field samples are identified by Zhu et al. (2019). The lab clonal samples include 5 lab clonal samples (7G8, GB4, KH02, KE01, GN01) and a number of crosses samples comprising parents and progeny from the *P. falciparum* Genetic Crosses project (Miles et al., 2016). The lab mixed samples include 27 mixed lab strains in varying proportions created by Dr Jason Wendler (Wendler, 2015).

S1.2 Generating *in silico* mixtures for deconvolution benchmark

S1.3 *In silico* mixtures of lab strains

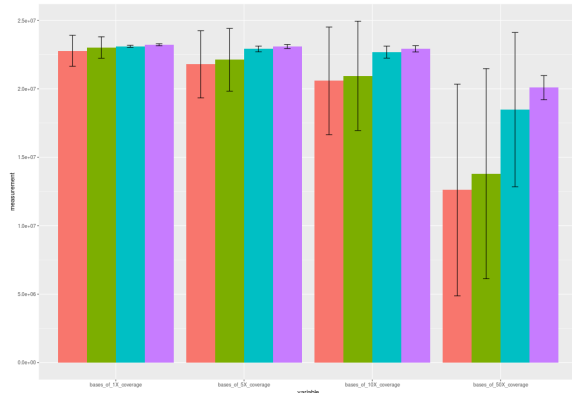
In addition to the *in vitro* lab mixtures from Wendler (2015), we generated four *in silico* lab mixtures. We used the lab strain GB4 to provide a whole-genome read-depth profile, and drew Bernoulli variables to simulate alternative allele counts, with the probability of success set to the adjusted within-sample allele frequency (WSAF) at each position. The expected WSAF is the dot product of the genotype of the lab strains (3D7, Dd2, HB3, 7G8) and the proportions. For example, if the genotype is 0, 0, 1, 1 at an arbitrary position, a mixture proportion of 0.1, 0.2, 0.4 and 0.3 leads to an expected WSAF of 0.7. We then adjust the WSAF with an error term (0.01) to account for technical error/noise: $0.7 \times (1 - 0.01) + (1 - 0.7) \times 0.01 = 0.696$.

S1.4 Data quality

S1.5 Variant selection

We ran DEploid-IBD on high-quality biallelic SNPs (both coding and non-coding, tagged with PASS at the QUAL column in the VCF file) from Pf3k (Pf3k Consortium, 2016). Before the additional filtering described below, this set contained 1,057,830 SNPs.

```
bcftools view \
--include 'FILTER="PASS"' \
--min-alleles 2 \
--max-alleles 2 \
--types snps \
--output-file SNP_INDEL_Pf3D7_01_v3.high_quality_biallelic_snps.vcf.gz \
--output-type z \
SNP_INDEL_Pf3D7_01_v3.combined.filtered.vcf.gz
```

A coverage**B VQSLOD**

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S1.6 Variant quality score log-odds

The variant quality score log-odds (VQSLOD) score measures how reliable each variant call is. For the Pf3k data set, all ‘PASS’ sites have a VQSLOD score above zero.

S1.7 Data cleaning and filtering

S1.7.1 High leverage data points

As in Zhu et al. (2018, 2019), we find that markers with high frequencies in both reference and alternative allele counts can mislead our model into fitting extra strains. We used a threshold of $\geq 99.5\%$ coverage (default) to identify markers with extremely high allele counts. We further expand this list of potential “bad markers” by considering their nearest 10 neighbours on both sides along the genome, and identify the nearest markers as “to be excluded” if overlaps are found (see Figure S1.1). These poorly-genotyped variants are likely to be artefacts of mapping and genotype calling.

This process remains a manual step, as it may consider other sites.

S1.8 Reference panel

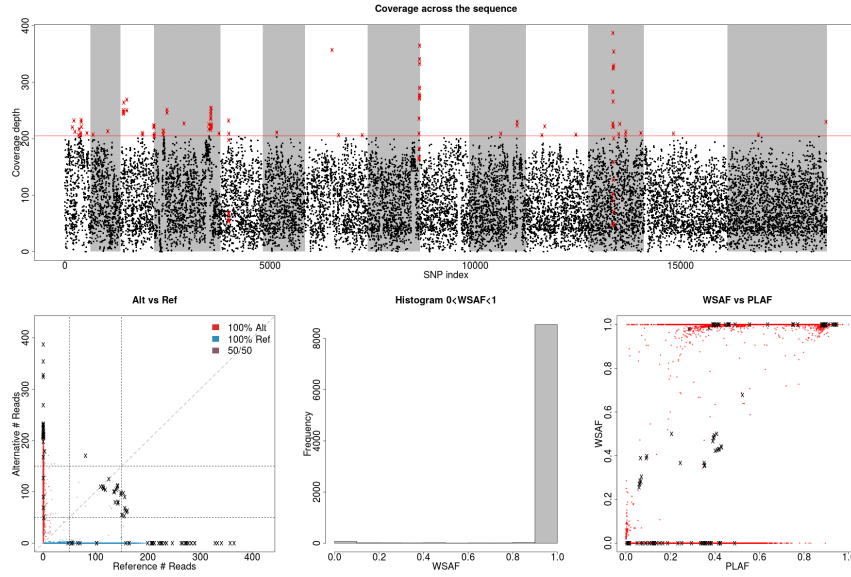


Figure S1.1: The top plot shows total allele counts across all markers. We observe a small number of heterozygous sites with high coverage (marked as crosses above), which can potentially mislead our model into over-fitting the data with additional strains. We use a threshold of $\geq 99.5\%$ coverage to identify markers with extremely high allele counts; red crosses indicate markers that are filtered out. The left plot shows the alternative allele count against the reference allele count. As *P. falciparum* genomes are haploid, in clonal samples one would expect to see either the alternative or the reference allele at any site; heterozygous sites are indications of mixed infection. The marked black crosses can mislead our model into overestimating the number of strains as two. The middle panel is a histogram of the within-sample allele frequency, and the right panel shows the within-sample allele frequency compared against the population average.

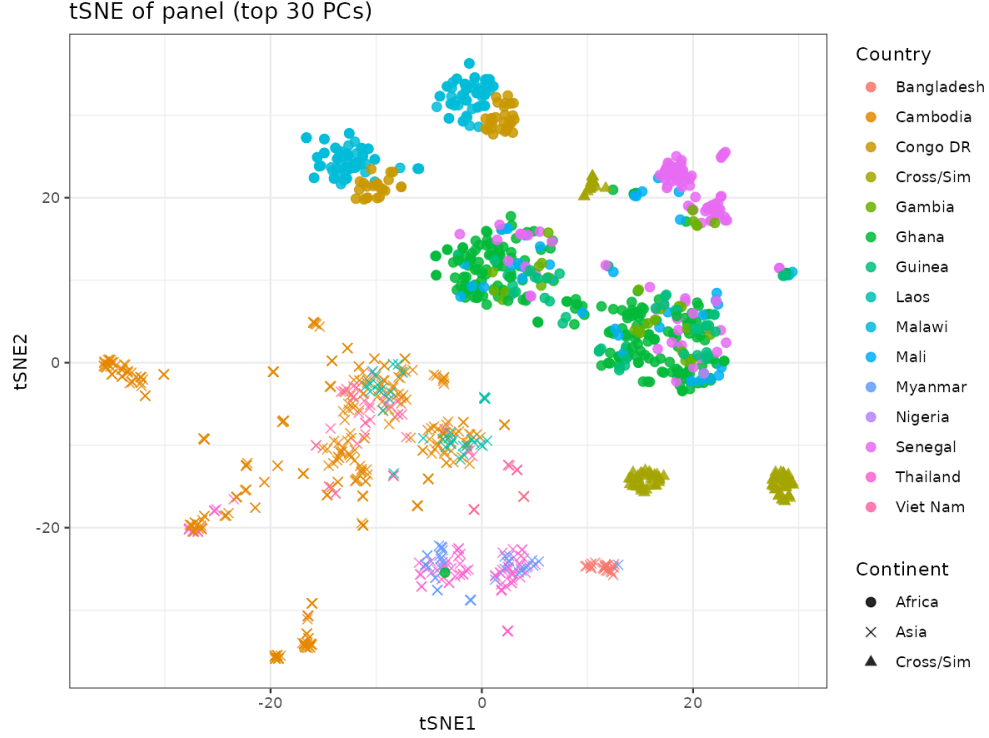


Figure S1.2: tSNE projection of the combined Pf3k field reference panel (18,570 sites $\times$ 1,316 haplotype samples), coloured by sample country of origin; lab genetic crosses and complexity-of-infection simulations (blank country in the Pf3k metadata) are grouped as Cross/Sim. The projection recovers the major geographic structure of the panel.

S2 DEploid-LASSO deep dive

S2.1 Model

We recall the notation of Zhu et al. (2019). Using a Bayesian approach, the posterior probability of the strain number K , proportions $\mathbf{w}$, haplotypes $\mathbf{h}$ and relatedness state $\mathcal{S}$ given the sequence data D and error rate e is

$$P(K, \mathbf{w}, \mathbf{h}, \mathcal{S} \mid e, D) \propto L(K, \mathbf{w}, \mathbf{h}, \mathcal{S} \mid e, D) \times P(K, \mathbf{w}, \mathbf{h}, \mathcal{S}), \quad (\text{S2.1})$$

where the joint prior decomposes as

$$P(K, \mathbf{w}, \mathbf{h}, \mathcal{S}) = P(K) \times P(\mathbf{w} \mid K) \times P(\mathcal{S} \mid K) \times P(\mathbf{h} \mid \mathcal{S}). \quad (\text{S2.2})$$

This decomposition allows us to be flexible with the order of inference. In practice we use Markov chain Monte Carlo (MCMC) for inference, and we find that certain execution orders perform better than others, as described in the main text.

S2.2 LASSO reference-panel optimisation

Let y denote the within-sample allele frequency of the mixed sample. Assuming that the genotypes of each strain in the reference panel X are independent covariates, we fit the linear model

$$y = \beta_0 + \vec{\beta} X, \quad (\text{S2.3})$$

where $\vec{\beta}$ are the weights of the reference strains that contribute towards the estimated within-sample allele frequency of the mixed genome. A key feature of the LASSO approach is that $\vec{\beta}$ is sparse, producing zero weights for many reference strains in the panel and thereby reducing its size. Our objective is to find $\vec{\beta}$ and λ such that

$$\min_{\vec{\beta}} \left\| y - \beta_0 - \vec{\beta} X \right\|^2 + \lambda \left\| \vec{\beta} \right\|_1 \quad (\text{S2.4})$$

(Friedman et al., 2010).

In practice we search for λ between 0 and 1. We start by setting $\lambda = 1$ and $\vec{\beta} = \mathbf{0}$, then gradually add reference strains into the model by setting individual β s to non-zero values one at a time. We keep including new reference strains if the new model reduces the likelihood deviance by at least one percent. The LASSO approach lets us discard reference strains with zero β s, and we further reduce the panel by opting out β s below 0.01. Note that the choice of the final reference strains depends on the order in which reference strains are included in the computation.

S2.3 Effective reference panel

The LASSO step yields a *sparse* weight vector $\vec{\beta}$; the effective reference panel is the subset of input strains with $\beta_j \geq 0.01$ (Section S2.2). This is the covariate set actually passed to the downstream IBD haplotype inference, rather than the full m -strain panel. Because a strain is only admitted when it reduces the deviance by at least 1%, the effective panel size is set by the number of distinct components in the mixture – typically a small multiple of K – and is essentially independent of the input panel size m . Two consequences follow. First, the downstream MCMC cost scales with the effective, not the full, panel size, which is why **DEploid-BEST** runs in roughly constant time as m grows from 10 to 1000 (Table S1). Second, by discarding redundant and related strains the effective panel is non-over-parameterised, allowing the strain number to be recovered automatically instead of collapsing related components as **DEploid-IBD** does (Tables S2–S3). The selection is performed per sample, since each mixture has its own composition.

S3 Major improvement from DEploid and DEploid-IBD

S3.1 Optimised reference panel per sample

S3.2 Reduced run time

To isolate the effect of reference-panel size on compute cost, we deconvolved a fixed set of $l = 18,570$ sites (sample PG0390-C) using synthetic reference panels of $m \in \{10, 20, 50, 100, 200, 500, 1000\}$ strains, holding the MCMC settings fixed ($nSample = 20$, $rate = 5$, seed 1, single thread). The classic DEploid and DEploid-IBD pipelines scale approximately quadratically with m and fail to complete within a 5-minute budget once m exceeds roughly 20–50 strains, whereas DEploid-BEST completes in roughly 2–3 minutes irrespective of panel size, including $m = 1000$ (Table S1).

Reference strains (m)	Runtime			Peak memory (GB)		
	classic	IBD	BEST	classic	IBD	BEST
10	13.1 s	1.4 m	2.0 m	0.06	0.06	0.07
20	2.1 m	2.8 m	2.0 m	0.06	0.06	0.07
50	TIMEOUT	TIMEOUT	2.5 m	0.10	0.10	0.09
100	TIMEOUT	–	2.4 m	0.19	–	0.10
200	–	–	2.4 m	–	–	0.14
500	–	–	2.5 m	–	–	0.27
1000	–	–	2.7 m	–	–	0.50

Table S3.2: Supplementary Table S1. Runtime and peak memory of the three DEploid approaches as a function of reference-panel size m ($l = 18,570$ sites, $nSample = 20$, $rate = 5$, seed 1, single thread). Runs exceeding 300 s were terminated (shown as TIMEOUT). Synthetic panels were used to vary m while holding the site set fixed; strain haplotypes were assigned at random (seed 42). Hardware: Intel Core i9-14900HX, 32 threads, 31 GB RAM.

S3.3 Accuracy on in-silico mixtures of known strains

To quantify inference accuracy we deconvolved four *in-silico* mixtures of four lab strains (3D7, DD2, HB3, 7G8) with known ground-truth proportions, including a balanced 0.25/0.25/0.25/0.25 mixture (Table S2). Each mixture was run through the classic DEploid pipeline, DEploid-IBD, and DEploid-BEST with a four-strain reference panel ($l = 18,570$ chromosome-14 sites). DEploid-BEST recovered the correct number of strains ($K = 4$) automatically in every case; the classic pipeline reached $K = 4$ only because K was supplied as input, while DEploid-IBD merged the two most closely related strain pairs and collapsed to $K = 2$ even when $K = 4$ was supplied. For proportion recovery, DEploid-BEST and the classic pipeline given the true K are statistically indistinguishable (mean total absolute error 0.35 for both; IBD 1.06). The residual error in both is dominated by a 3D7/7G8 labelling ambiguity: the two most divergent strains are exchanged for the largest proportion (e.g. for w1 the methods recover $3D7 \approx 0.39$, $7G8 \approx 0.21$ versus truth 0.1/0.4), an identifiability limit of the reference panel rather than a method failure. The practical benefit of DEploid-BEST on these mixtures is therefore its automatic, robust strain-number selection and its avoidance of the related-strain collapse seen in DEploid-IBD, together with the runtime scaling reported in Table S1.

S3.4 Accuracy on in-silico mixtures of field strains

To test the method on realistic data we built *in-silico* mixtures from genuine *P. falciparum* field haplotypes rather than the four reference lab strains. For each of two field populations (asiaGroup1, africaGroup2) we sampled a 20-strain reference panel and two further field strains whose haplotypes were tiled into mosaic “truth” genomes with one to four break points (the 1–4 “bites” scenarios); read counts were then simulated

Mixture	Method	K	Recovered proportion				$ \Delta $
			3D7	dd2	HB3	7G8	
w1	truth	4	0.10	0.20	0.30	0.40	–
	classic	4	0.39	0.10	0.30	0.20	0.590
	IBD	2	0.71	0.00	0.29	0.00	1.228
	BEST	4	0.38	0.11	0.30	0.21	0.569
w2	truth	4	0.25	0.25	0.25	0.25	–
	classic	4	0.25	0.25	0.25	0.25	0.005
	IBD	2	0.72	0.00	0.28	0.00	1.000
	BEST	4	0.25	0.25	0.25	0.25	0.008
w3	truth	4	0.20	0.20	0.20	0.40	–
	classic	4	0.40	0.20	0.20	0.20	0.400
	IBD	2	0.73	0.00	0.27	0.00	1.200
	BEST	4	0.40	0.20	0.20	0.19	0.411
w4	truth	4	0.30	0.30	0.30	0.10	–
	classic	4	0.30	0.10	0.30	0.30	0.402
	IBD	2	0.70	0.00	0.30	0.00	0.800
	BEST	4	0.30	0.10	0.30	0.30	0.410

Table S3.3: Supplementary Table S2. Recovered number of strains (K) and strain proportions for four synthetic mixtures of four lab strains (3D7, DD2, HB3, 7G8) with known truth. **DEploid-BEST** recovers the correct $K = 4$ automatically in every case; the classic pipeline only reaches $K = 4$ because K was supplied, and **DEploid-IBD** collapses to $K = 2$ even when $K = 4$ is supplied (it merges the two most closely related strain pairs). **DEploid-BEST** and the classic pipeline given the true K recover proportions to statistically indistinguishable accuracy (mean $|\Delta| = 0.35$ for both; IBD 1.06); the residual error is a 3D7/7G8 labelling ambiguity among the closely related reference strains. $|\Delta|$ is the total absolute error across the four proportions. Settings: single seed, $nSample = 500$, $rate = 8$, 4-strain reference panel, $l = 18,570$ chromosome-14 sites (matching the original benchmark fidelity; the original used 30 seeds).

at the observed field coverage by binomial sampling (error rate 0.01). The true proportions of the four mosaic components are encoded in the mixture name (e.g. **p25v25v25v25** denotes a balanced 0.25/0.25/0.25/0.25 mix). Because the true strains are *not* members of the reference panel, the deconvolution cannot be scored by matching recovered strain proportions; instead we score the recovered *mixture allele frequency* (WSAF) site by site, correlating the inferred WSAF ($\sum_j h_{sj} w_j$ over the recovered haplotypes h and proportions w) against the simulated truth. Each of the 40 mixtures ($2 \times 5 \times 4$) was deconvolved by the classic pipeline, **DEploid-IBD**, and **DEploid-BEST** (single seed, $nSample = 500$, $rate = 8$).

Table S3 reports the per-site WSAF recovery. All three pipelines recover the mixture allele frequency with very high accuracy (Pearson r in the range 0.97–0.99, typically ≈ 0.99 , in every setting): with a dense, representative field panel the mixture is identifiable regardless of strain-number misspecification. **DEploid-BEST** attains this accuracy *without* K being supplied – the classic and IBD pipelines were both fixed at the true $K = 4$ – and its recovery is if anything marginally better than the fixed- K pipelines in every scenario. **DEploid-BEST** automatically selects a strain number (mean $K \approx 2.7$ across the 40 mixtures, ranging 2.1–3.3 by scenario, versus the simulated 4; Table S3, last column), under-estimating the true count by merging the most similar components. Crucially, it does so without K being supplied, and this under-estimate does not degrade recovery: the mixture allele frequency is still recovered to $r \approx 0.99$ (Table S3), matching the fixed- $K = 4$ classic and IBD pipelines. Together with the lab-strain benchmark (Table S2) and the runtime scaling (Table S1), this shows that on realistic field mixtures **DEploid-BEST** is as accurate as a correctly specified classical deconvolution while removing the need to choose K by hand.

Table S3.4: Supplementary Table S3. Field in-silico benchmark: per-site WSAF recovery. For 40 simulated mixtures (two field populations $\times$ five mosaic scenarios $\times$ four known proportion sets), each pipeline deconvolves the mixture and we correlate the recovered mixture allele frequency (WSAF) against the simulated truth, site by site. Means are over both populations and all four proportion sets; the range is in parentheses. DEploid-BEST requires no strain number K to be supplied. Settings: single seed, $nSample = 500$, $rate = 8$, matching the original field-benchmark fidelity (Table S2 uses the same $nSample$ and $rate$).

Scenario	DEploid-BEST	classic (fixed $K=4$)	DEploid-IBD (fixed $K=4$)	BEST K
1	0.990 (0.985–0.996)	0.988 (0.984–0.992)	0.990 (0.986–0.994)	2.12
2a	0.988 (0.986–0.994)	0.987 (0.980–0.993)	0.985 (0.977–0.990)	2.75
2b	0.989	0.987 (0.982–0.993)	0.988 (0.983–0.991)	2.50
3	0.986 (0.983–0.992)	0.985 (0.978–0.990)	0.985 (0.979–0.990)	3.00
4	0.984 (0.980–0.989)	0.983 (0.975–0.990)	0.984 (0.978–0.991)	3.25

S4 Modularised package development

To facilitate the development and maintenance of the DEploid software, we have modularised the package into several components. Each component is responsible for a specific functionality, making the codebase more organised and easier to manage. The main modules are:

- **DEploid-Lasso-lib (C++)**: Handles the input and preprocessing of genomic data.
- **DEploid-vcf-lib (C++)**: Contains the core algorithms for deconvolution and inference.
- **DEploid.utils (R)**: Utility functions for data preprocessing.
- **DEploid (C++)**: DEploid core functions, command-line interface.
- **DEploid-r (R)**: An R interface for DEploid, allowing users to interact directly with DEploid results without saving them to text files and reloading them for analysis. Upstream direct dependencies include DEploid and DEploid.utils.

This modular approach allows for independent development and testing of each component, improving the overall robustness and flexibility of the software.

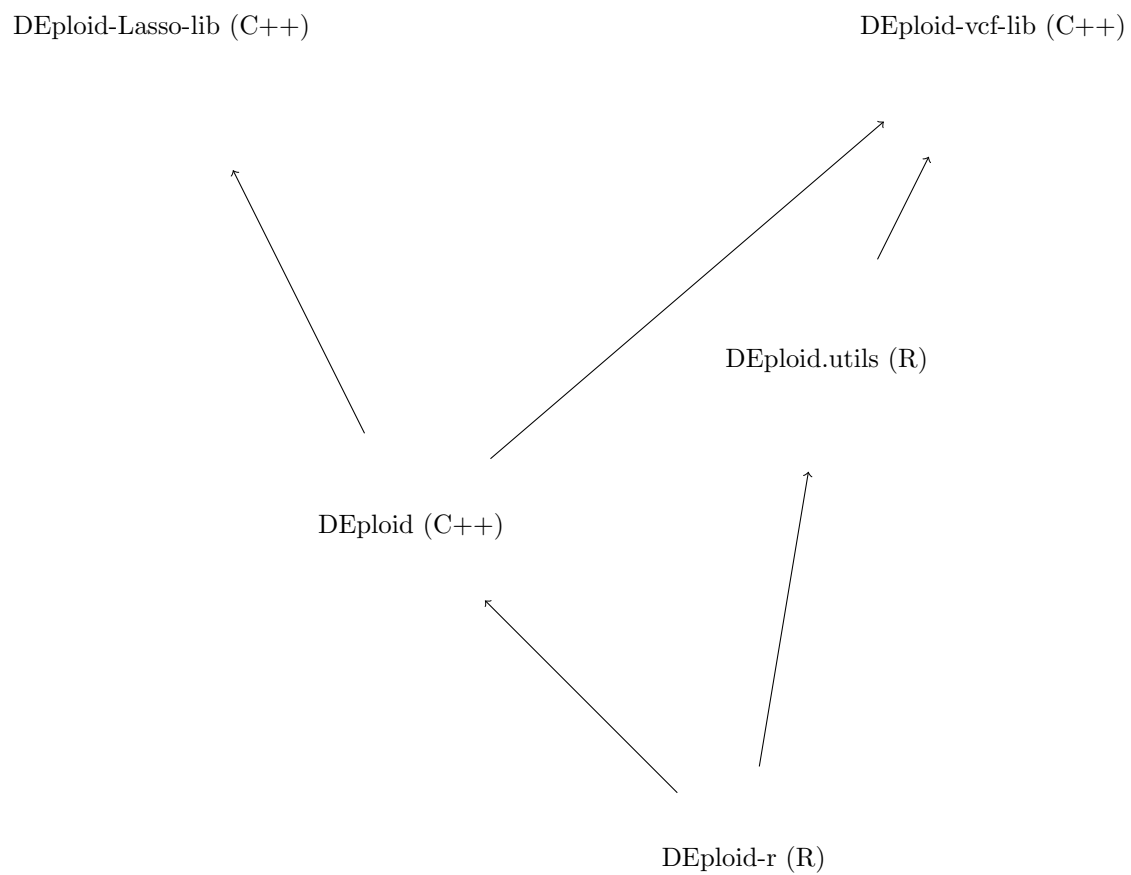


Figure S4.3: Package dependencies.

S5 Deconvolution of the field samples

Unlike the benchmark mixtures in Sections S2-S3, here we deconvolve whole-genome data from the field samples with the production **DEploid-BEST** pipeline. This section documents the input data, the end-to-end workflow, and the measured compute cost of deconvolving all 5,970 samples of the Pf6 analysis set.

S5.1 Input data and reference panel

Application of **DEploid-BEST** to the complete *Pf6* analysis set (**pf6-analysis-meta-with-k-relatedness-collapsed-point**) illustrates the scale of the task. The set contains **5,970** samples, of which 533 are flagged **collapsed** (redundant or related samples merged), leaving $\approx$ **5,437** unique individuals. The relatedness-derived strain-count estimate **Est_K** classifies them as **3,434** clonal infections (**Est_K** = 1) and **2,536** mixed infections (**Est_K** $\geq$ 2: 1,701 at *K*=2, 658 at *K*=3, 177 at *K*=4). Only the mixed infections require full deconvolution; the clonal samples resolve to a single strain and are substantially cheaper to process.

The deconvolution takes two inputs per sample:

- **Mixture VCF.** Each *Pf6* sample is extracted to a single-sample whole-genome VCF carrying that sample's allele-depth (AD) counts at every variant site (**data/pf6.vcf/by_sample/<SAMPLE>.vcf.gz**; 5,970 files).
- **Reference panel.** A subset of the combined *Pf3k* field panel (**combined_pf3k_field_panel_withcross.txt**) restricted to the sites shared with the *Pf6* calls: **1,316** strains $\times$ **15,052** biallelic SNP positions (**panel.subset.txt**).

dEploid requires the panel and the input VCF to contain exactly the same site set, so each sample VCF is intersected to the 15,052 panel positions before inference. Population allele frequencies are read directly from the VCF via **-plafFromVcf** (the **AF INFO** tag, computed as **AC/AN** across all 7,113 *Pf6* samples), which guarantees the **PLAF** matches the VCF's site set.

S5.2 Workflow

The pipeline (**run_all.sh**) is organised into five stages, summarised in Figure S5.4:

- Stage 1 Panel subsetting & variant selection** (**make_panel_vcf.sh**). Intersect the *Pf3k* reference panel with the *Pf6* variant calls and retain high-quality biallelic SNPs, yielding the $1,316 \times 15,052$ **panel.subset.txt**.
- Stage 2 Per-sample VCF preparation** (**add_af.sh**). For every sample VCF, attach the population **AF** = **AC/AN** tag (required by **-plafFromVcf**) and subset the records to the 15,052 panel positions.
- Stage 3 LASSO panel optimisation & deconvolution** (**deconvolve_one.sh** $\rightarrow$ **dEploid -best**). Within **dEploid -best** the effective panel is first shrunk by LASSO (Section S2.2), then **chooseK** selects the strain number *K*, **DEploid-IBD** reconciles the haplotypes and proportions, and the final haplotypes are written to **<S>.final.hap**.
- Stage 4 Per-sample summary table.** **deconvolve_one.sh** parses *K* and the column-wise mean IBD proportions and writes out **<S>.txt** with the header **# sample <S>**, mixed of *K* strain, with proportions of **p1, p2, ...** followed by the **CHROM/POS/h₁...h_K** haplotype table.
- Stage 5 Archiving** (**make_archive.sh**). All **out/*.txt** files are zipped into **deconvolved_haplotypes.zip**.

S5.3 Parallel execution and outputs

The 5,970 samples are independent, so `run.all.sh` launches one `deconvolve.one.sh` per sample and runs them `NCORES` at a time (`xargs -P 20`; GNU parallel is used when available). The driver is resume-safe: a sample whose `out/<S>.txt` already exists is skipped, so an interrupted run can be restarted without redoing completed work. On completion, `make_archive.sh` bundles every per-sample table into a single `deconvolved.haplotypes.zip`. As of this writing the run had produced **645** per-sample tables; a preliminary read of their deconvolved K gives 421 clonal ($K=1$), 178 mixed ($K=2$) and 44 mixed ($K=3$) samples, consistent with the `Est_K` expectation above.

S5.4 Compute time and throughput

The run used 20 parallel single-threaded `dEplod` workers on one machine, with the production MCMC settings $nSample = 250$, $rate = 8$, $burn = 0.67$, seed 1. It sustained a throughput of ≈ 4 samples per minute across the 20 workers, i.e. ≈ 5 CPU-minutes per sample (Table S4). At this rate the full 5,970-sample set is projected to finish in roughly 25 hours of wall-clock time on 20 cores – about 21 single-core days of compute. This is substantially faster than the ≈ 10 single-core days implied by the synthetic-panel Table S1 rate, because the production panel is LASSO-reduced per sample (so the downstream MCMC sees only the few strains that contribute) and the wall time is divided across 20 cores. The speed-up is not perfectly linear, because the LASSO panel-optimisation stage is performed for every sample and $K=3/K=4$ samples take longer than clonal ones.

Quantity	Value
Samples deconvolved	5,970
Parallel <code>dEplod</code> workers	20 (single-threaded each)
MCMC settings	$nSample = 250$, $rate = 8$, $burn = 0.67$, seed 1
Sites per sample	15,052
Reference panel (input)	1,316 strains
Measured throughput	≈ 4 samples / min (20 cores)
Per-sample cost	≈ 5 CPU-min
Wall-clock on 20 cores	≈ 25 h (projected)
Single-core equivalent	≈ 21 days (projected)

Table S5.5: Supplementary Table S4. Measured compute cost of the Pf6 field-sample deconvolution with `DEplod-BEST`. Throughput is the sustained rate observed over the production run (645 of 5,970 samples completed at the time of writing); per-sample cost is derived as 20 cores divided by that throughput. The run is resume-safe, so the wall-clock total is unchanged by interruptions.

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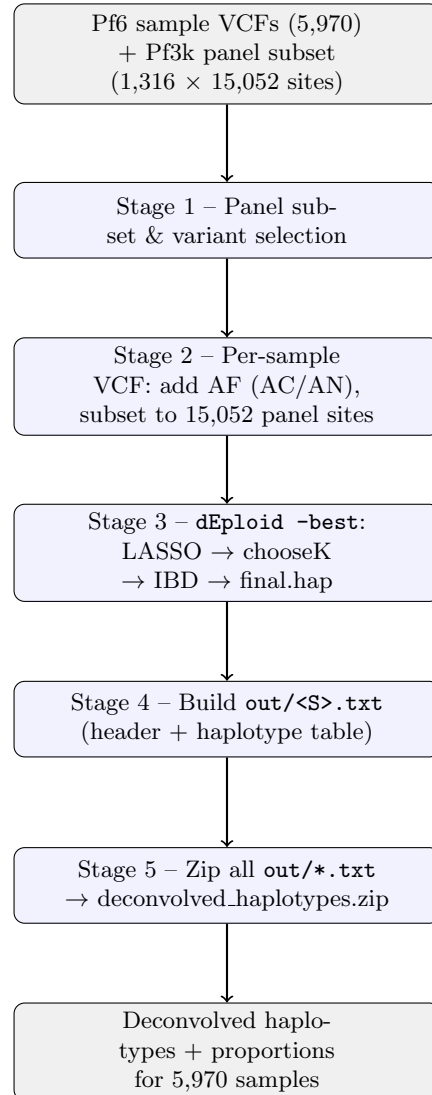


Figure S5.4: End-to-end DEploid-BEST field-sample deconvolution workflow. Each Pf6 sample VCF is intersected to the 15,052 panel sites and annotated with population allele frequency (AC/AN), then deconvolved by dEploid -best (LASSO panel shrinkage → strain-number selection → IBD haplotype/proportion inference → final haplotypes). Per-sample jobs run in parallel (`xargs -P 20`) and the resulting haplotype tables are archived.



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Figure S5.5: PCA